Practical Demo: Hedging a Bond Portfolio with Treasury Futures

Table of contents

Learning Objectives

After completing this demo you should be able to:

- calculate a bond portfolio's analytical and Treasury-beta-adjusted DV01;
- derive Treasury futures DV01 from the cheapest-to-deliver (CTD) bond and conversion factor;
- size and round a short futures hedge;
- compare portfolio P&L before and after the hedge; and
- identify residual basis, spread, curve, rounding, and CTD risks.

1 Market Inputs Versus Illustrative Inputs

The workflow resembles a daily risk-management task, but every number below is illustrative rather than live market data. A live hedge would use current positions, clean prices, accrued interest, key-rate durations, deliverable-basket data, conversion factors, CTD candidates, financing rates, and exchange contract specifications.

Simplifying assumptions:

- changes are small enough for first-order DV01 approximations;
- the selected CTD remains unchanged;
- futures DV01 is constant over the scenario;
- the Treasury move is parallel;
- spread duration equals modified duration for the corporate positions;
- yield betas summarize how each position historically responds to Treasury yields; and
- transaction costs, margin cash flows, convexity, carry, and roll are ignored.

2 Build the Portfolio Risk Table

The portfolio contains Treasuries and investment-grade corporate bonds. Analytical DV01 is

$$DV01_i = D_{\text{mod},i} \times MV_i \times 0.0001.$$

For cross-hedging, a Treasury yield beta maps each bond's historical price sensitivity to the hedging instrument:

$$DV01_i^{\text{Treasury}} = DV01_i \times \beta_i.$$

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

portfolio = pd.DataFrame({
    "Position": [
        "5-year Treasury notes",
        "10-year Treasury notes",
        "A industrial bonds",
        "BBB financial bonds",
    ],
    "Market value ($mm)": [30.0, 25.0, 24.0, 18.0],
```

```
    "Modified duration": [4.5, 7.6, 6.1, 7.2],
    "Treasury yield beta": [1.00, 1.00, 0.88, 0.72],
    "Spread shock in scenario (bp)": [0.0, 0.0, 12.0, 20.0],
})

portfolio["Analytical DV01 ($/bp)"] = (
    portfolio["Market value ($mm)"] * 1_000_000
    * portfolio["Modified duration"] * 0.0001
)
portfolio["Treasury DV01 ($/bp)"] = (
    portfolio["Analytical DV01 ($/bp)"] * portfolio["Treasury yield beta"]
)
portfolio["Spread DV01 ($/bp)"] = np.where(
    portfolio["Position"].str.contains("bonds"),
    portfolio["Analytical DV01 ($/bp)"],
    0.0,
)

portfolio.round(2)
```

	Position	Market value ($mm)	Modified duration	Treasury yield beta	Spread shock in
0	5-year Treasury notes	30.0	4.5	1.00	0.0
1	10-year Treasury notes	25.0	7.6	1.00	0.0
2	A industrial bonds	24.0	6.1	0.88	12.0
3	BBB financial bonds	18.0	7.2	0.72	20.0

```
analytical_dv01 = portfolio["Analytical DV01 ($/bp)"].sum()
treasury_dv01 = portfolio["Treasury DV01 ($/bp)"].sum()
spread_dv01 = portfolio["Spread DV01 ($/bp)"].sum()

pd.Series({
    "Portfolio market value ($mm)": portfolio["Market value ($mm)"].sum(),
    "Analytical DV01 ($/bp)": analytical_dv01,
    "Treasury-beta-adjusted DV01 ($/bp)": treasury_dv01,
    "Corporate spread DV01 ($/bp)": spread_dv01,
}).round(2)
```

```
Portfolio market value ($mm)              97.0
Analytical DV01 ($/bp)                 60100.0
```

```
Treasury-beta-adjusted DV01 ($/bp)    54714.4
Corporate spread DV01 ($/bp)          27600.0
dtype: float64
```

The analytical DV01 describes a common change in every position's discount yield. The beta-adjusted DV01 is the exposure targeted by a Treasury futures hedge. Spread risk remains a separate exposure.

3 Frame the CTD and Conversion-Factor Problem

Treasury futures are linked to a delivery basket. The short chooses the eligible security to deliver, and the cheapest-to-deliver issue often determines the futures contract's effective interest-rate sensitivity.

Suppose the current CTD candidate has the following illustrative inputs:

Input	Value
CTD clean price	101.75
CTD modified duration	7.35
Deliverable face amount	\$100,000
Conversion factor	0.9275
Futures price	112-16 = 112.50

Ignoring accrued interest and delivery-option details,

$$DV01_{CTD} = D_{\text{mod,CTD}} P_{CTD} (0.0001),$$

and

$$DV01_{futures} \approx \frac{DV01_{CTD}}{CF}.$$

```python
ctd_clean_price = 101.75
ctd_modified_duration = 7.35
deliverable_face = 100_000
conversion_factor = 0.9275
futures_price = 112.50

ctd_market_value = ctd_clean_price / 100 * deliverable_face
ctd_dv01 = ctd_modified_duration * ctd_market_value * 0.0001
```

```
futures_dv01 = ctd_dv01 / conversion_factor
approx_invoice_price = futures_price / 100 * deliverable_face * conversion_factor

pd.Series({
    "CTD market value ($)": ctd_market_value,
    "Approximate invoice price before accrued interest ($)": approx_invoice_price,
    "CTD DV01 ($/bp)": ctd_dv01,
    "Futures DV01 ($/bp)": futures_dv01,
}).round(2)
```

```
CTD market value ($)                                     101750.00
Approximate invoice price before accrued interest ($)    104343.75
CTD DV01 ($/bp)                                              74.79
Futures DV01 ($/bp)                                          80.63
dtype: float64
```

The futures contract does not have a true bond duration. Its practical sensitivity is inherited from the current CTD bond and scaled by the conversion factor.

4 Size and Round the Hedge

The portfolio is long bonds and loses when rates rise, so the hedge shorts Treasury futures. The exact contract count is

$$N^* = \frac{DV01^{\text{Treasury}}_{portfolio}}{DV01_{futures}}.$$

```
exact_contracts = treasury_dv01 / futures_dv01
rounded_contracts = int(np.rint(exact_contracts))
hedge_dv01 = rounded_contracts * futures_dv01
residual_treasury_dv01 = treasury_dv01 - hedge_dv01
hedge_ratio = hedge_dv01 / treasury_dv01

pd.Series({
    "Exact contracts to short": exact_contracts,
    "Rounded contracts to short": rounded_contracts,
    "Hedge DV01 ($/bp)": hedge_dv01,
    "Residual Treasury DV01 ($/bp)": residual_treasury_dv01,
    "Hedge ratio": hedge_ratio,
}).round(4)
```

```
Exact contracts to short             678.5687
Rounded contracts to short           679.0000
Hedge DV01 ($/bp)                  54749.1792
Residual Treasury DV01 ($/bp)        -34.7792
Hedge ratio                            1.0006
dtype: float64
```

Rounding creates a small residual exposure. A negative residual means the rounded hedge is slightly too large; a positive residual means it is slightly too small.

5 Before-and-After Scenario P&L

Suppose Treasury yields rise by 35 basis points. Corporate spreads also widen by the amounts in the portfolio table. The first-order bond P&L is separated into Treasury and spread components:

$$\Delta V_{portfolio} \approx -DV01^{\text{Treasury}} \Delta y_{Treasury} - DV01^{\text{spread}} \Delta s.$$

Because the hedge is short futures, its approximate gain when Treasury yields rise is

$$\Delta V_{futures} \approx N \times DV01_{futures} \times \Delta y_{Treasury}.$$

```
treasury_shock_bp = 35.0

portfolio["Treasury P&L ($)"] = -portfolio["Treasury DV01 ($/bp)"] * treasury_shock_bp
portfolio["Spread P&L ($)"] = (
    -portfolio["Spread DV01 ($/bp)"]
    * portfolio["Spread shock in scenario (bp)"]
)
portfolio["Total unhedged P&L ($)"] = (
    portfolio["Treasury P&L ($)"] + portfolio["Spread P&L ($)"]
)

futures_pnl = rounded_contracts * futures_dv01 * treasury_shock_bp
unhedged_pnl = portfolio["Total unhedged P&L ($)"].sum()
hedged_pnl = unhedged_pnl + futures_pnl
residual_rate_pnl = -residual_treasury_dv01 * treasury_shock_bp
spread_pnl = portfolio["Spread P&L ($)"].sum()

portfolio[[
```

```
    "Position", "Treasury P&L ($)", "Spread P&L ($)", "Total unhedged P&L ($)"
]].round(0)
```

	Position	Treasury P&L ($)	Spread P&L ($)	Total unhedged P&L ($)
0	5-year Treasury notes	-472500.0	-0.0	-472500.0
1	10-year Treasury notes	-665000.0	-0.0	-665000.0
2	A industrial bonds	-450912.0	-175680.0	-626592.0
3	BBB financial bonds	-326592.0	-259200.0	-585792.0

```
pd.Series({
    "Unhedged portfolio P&L ($)": unhedged_pnl,
    "Short futures P&L ($)": futures_pnl,
    "Hedged portfolio P&L ($)": hedged_pnl,
    "Residual rate P&L from rounding ($)": residual_rate_pnl,
    "Unhedged spread P&L ($)": spread_pnl,
}).round(0)
```

```
Unhedged portfolio P&L ($)             -2349884.0
Short futures P&L ($)                   1916221.0
Hedged portfolio P&L ($)                -433663.0
Residual rate P&L from rounding ($)        1217.0
Unhedged spread P&L ($)                 -434880.0
dtype: float64
```

The futures hedge removes almost all of the modeled Treasury-rate loss. It does not hedge the corporate spread widening, which remains the dominant loss after the hedge.

```
treasury_scenarios = np.arange(-75, 76, 5)
unhedged_rate_pnl = -treasury_dv01 * treasury_scenarios
hedged_rate_pnl = unhedged_rate_pnl + hedge_dv01 * treasury_scenarios

fig, ax = plt.subplots(figsize=(7.5, 4.1))
ax.plot(treasury_scenarios, unhedged_rate_pnl / 1_000, linewidth=2.1,
        label="Unhedged Treasury-rate P&L")
ax.plot(treasury_scenarios, hedged_rate_pnl / 1_000, linewidth=2.1,
        label="After rounded futures hedge")
ax.axhline(0, color="black", linewidth=0.8)
ax.axvline(0, color="gray", linestyle="--", linewidth=1)
ax.set(xlabel="Parallel Treasury yield shock (bp)", ylabel="Rate-driven P&L ($ thousands)",
```

```
       title="First-order Treasury-rate hedge")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()
```

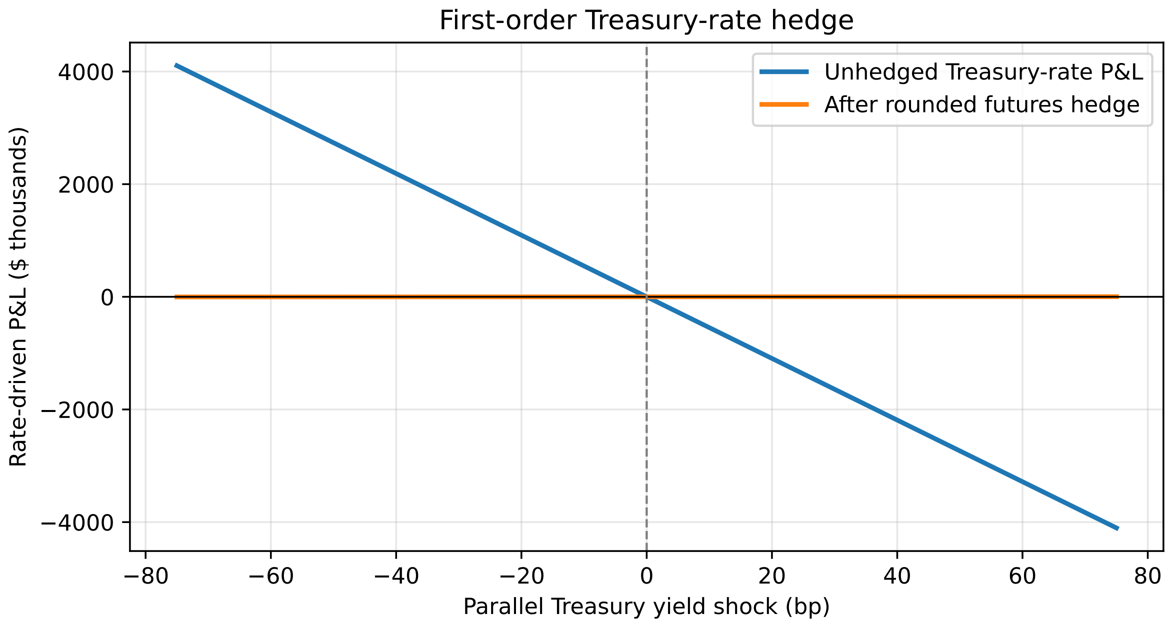

Figure 1: The futures hedge offsets parallel Treasury-rate risk but not spread risk

6 Validation Checks

```
assert ctd_dv01 > 0 and futures_dv01 > 0
assert rounded_contracts > 0
assert abs(treasury_dv01 - hedge_dv01 - residual_treasury_dv01) < 1e-8
assert abs(hedged_pnl - (spread_pnl + residual_rate_pnl)) < 1e-8
assert abs(residual_treasury_dv01) <= 0.5 * futures_dv01 + 1e-10

checks = pd.Series({
    "Futures DV01 is positive": futures_dv01 > 0,
    "Long-bond hedge requires short futures": rounded_contracts > 0,
    "Rounded residual is no more than half one contract DV01": (
```

```
        abs(residual_treasury_dv01) <= 0.5 * futures_dv01
    ),
    "Hedged P&L reconciles to spread plus residual-rate P&L": (
        abs(hedged_pnl - spread_pnl - residual_rate_pnl) < 1e-8
    ),
})
checks
```

```
Futures DV01 is positive                                   True
Long-bond hedge requires short futures                     True
Rounded residual is no more than half one contract DV01    True
Hedged P&L reconciles to spread plus residual-rate P&L     True
dtype: bool
```

7 What Can Break the Hedge?

- **Basis risk:** cash bonds and the futures contract may not move together.
- **Curve risk:** one CTD-based DV01 cannot hedge every key-rate exposure.
- **Spread risk:** Treasury futures do not directly hedge corporate credit spreads.
- **Yield-beta error:** historical betas can change in stress periods.
- **CTD switching:** a different deliverable bond can become cheapest to deliver.
- **Conversion-factor and delivery-option effects:** the simple DV01 scaling is approximate.
- **Convexity:** large yield moves make a first-order DV01 hedge less accurate.
- **Rounding, liquidity, margin, and roll:** implementation affects realized results.

8 Financial Interpretation

A Treasury futures hedge is a risk-matching exercise, not a notional-matching exercise. The correct sequence is:

1. identify the portfolio risk to hedge;
2. map that risk into Treasury DV01;
3. estimate futures DV01 from the current CTD and conversion factor;
4. choose and round the contract count; and
5. test the combined position under rate, spread, curve, and basis scenarios.

The hedge is successful here because it neutralizes the stated parallel Treasury-rate exposure. The remaining loss is economically meaningful: it is primarily credit spread risk, which was never within the hedge's scope.